Nonfreeness for rational two-parabolic groups with numerator 24

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Abstract

Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $B_q = \left(\begin{smallmatrix} 1 & q \\ 0 & 1 \end{smallmatrix}\right)$. We prove that $\langle A, B_{24/r} \rangle \leq \mathrm{SL}_2(\mathbb{C})$ is not a free group for every integer r with $|r| > 6$. Each parameter has a nontrivial relation of at most 64 syllables. Moreover, $24/r$ is a two-step relation number whenever $|r| > 17$. The proof combines a finite certificate of 48 good residue-class rules with a five-syllable identity for every nonzero integer x and $R = 1 + 1260x$:

$$B_{24/R}^{3Rx} A^{18+630x} B_{24/R}^{-3} A^{18x} B_{24/R}^{R(3+105x)} = \begin{pmatrix} 1 & 0 \\ 18/R & 1 \end{pmatrix}.$$

This identity covers the four residue classes left by the finite certificate. All finite data and their exact verification are included, and the accompanying Lean project proves the universal nonfreeness and bounded-relation statements.

1 Introduction

For $q \in \mathbb{C}$, put

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad B_q = \begin{pmatrix} 1 & q \\ 0 & 1 \end{pmatrix}, \qquad G_q = \langle A, B_q \rangle \leq \mathrm{SL}_2(\mathbb{C}). \tag{1.1}$$

The rational two-parabolic problem asks whether G_q is nonfree for every nonzero rational q with $|q| < 4$. Kim and Koberda [1, Theorem 1.4] proved this for parameters $q = s/r$ with $0 < |s| \leq 27$ and $|s| \neq 24$. For numerator 24, they proved that the denominators giving two-step relation numbers have natural density one. We prove the all-denominator conclusion for this numerator.

A *syllable* is a nonzero integer power of one formal generator. The syllable length of a freely reduced word counts its alternating power blocks; it does not count the letters obtained by expanding those powers. Let $F(a, b)$ denote the free group on a, b, and let $\mathrm{ev}_q : F(a, b) \to \mathrm{SL}_2(\mathbb{C})$ send a to A and b to B_q.

Theorem 1.1. *For every integer r satisfying $|r| > 6$, the group $G_{24/r}$ is not an abstract free group. There is a nonempty freely reduced word $v_r \in F(a, b)$ with at most 64 syllables such that*

$$v_r \neq 1, \qquad \mathrm{ev}_{24/r}(v_r) = I.$$

No coprimality assumption on r and 24 is required.

Following [1, Definition in Section 1], a nonzero parameter q is a *two-step relation number* if an alternating word

$$b^{e_1}a^{e_2}\cdots b^{e_{2k+1}}, \qquad 0 \le k \le 2, \qquad e_j \in \mathbb{Z}\setminus\{0\}, \tag{1.2}$$

has a lower-triangular value under ev_q.

Corollary 1.2. *The parameter $24/r$ is a two-step relation number for every integer r with $|r| > 17$. In particular, the set of such denominators is cofinite in $\mathbb{Z}\setminus\{0\}$.*

Corollary 1.3. *If $s, r \in \mathbb{Z}$, $r \neq 0$, $0 < |s| \le 27$, and $|s/r| < 4$, then $G_{s/r}$ is not a free group.*

Proof. Use Theorem 1.1 when $|s| = 24$, and [1, Theorem 1.4(1)] otherwise. The identity $B_{-q} = B_q^{-1}$ handles the signs. □

The last corollary uses the existing results for other numerators. The argument below is self-contained for numerator 24; it makes no claim for all rational parameters in $(-4, 4)$.

The mechanism of the proof

The good-class criterion of Kim and Koberda reduces many parameters to a five-syllable lower-unipotent matrix. We use 48 explicit rules, including rules for denominators that are not coprime to 24. Together they cover 40316 of the 40320 residue classes modulo 40320. The missing representatives are

$$1261, \quad 6299, \quad 34021, \quad 39059. \tag{1.3}$$

Each is congruent to 1 or -1 modulo 1260. The same five-syllable mechanism admits a particularly simple specialization on these two progressions: set $R = 1 + 1260x$ and use

$$m = 18x, \quad y = 3x, \quad u = 1 + 36x, \quad t = 3, \quad e = -3 - 105x.$$

Then $e = y - ut$ and $Ru + 24me = 1$. These two equations give the identity in the abstract, with no recursive descent or valuation calculation. The finitely many exceptions to the good-class rules are handled by 23 exact words.

The logical roles of the ingredients are separate. The residue-class check is a complete finite computation, the parametric identity handles every integer in the remaining infinite families, and free reduction proves that the resulting relations are nontrivial. We also give the group-theoretic argument that a nontrivial relation for these two noncommuting generators implies abstract nonfreeness. Section 6 identifies the corresponding Lean statements and the verification boundary.

2 A five-syllable matrix identity

Write $L(z) = \left(\begin{smallmatrix} 1 & 0 \\ z & 1 \end{smallmatrix}\right)$ and $U(z) = \left(\begin{smallmatrix} 1 & z \\ 0 & 1 \end{smallmatrix}\right)$. For every integer n,

$$A^n = L(n), \qquad B_q^n = U(nq). \tag{2.1}$$

Our commutator convention is $[X, Y] = XYX^{-1}Y^{-1}$.

Lemma 2.1. *Let $s, r, m, y, u, t, e, k \in \mathbb{Z}$, with $r \neq 0$, satisfy*

$$m = ky, \qquad e = y - ut, \qquad ru + sme = 1. \tag{2.2}$$

For $B = B_{s/r}$, one has

$$W := B^{ry} A^{-ke} B^{-t} A^{m} B^{-re} = L(kt/r). \tag{2.3}$$

If m, y, t, e are all nonzero, the formal word w displayed in (2.3) has five nonzero syllables, and $[w, a]$ is a nontrivial relation of exactly 12 syllables.

Proof. Equations (2.1) and (2.2) give

$$B^{ry} A^{-ke} = \begin{pmatrix} 1 - skye & sy \\ -ke & 1 \end{pmatrix} = \begin{pmatrix} ru & sy \\ -ke & 1 \end{pmatrix},$$
$$B^{ry} A^{-ke} B^{-t} = \begin{pmatrix} ru & se \\ -ke & 1 + sket/r \end{pmatrix}.$$

After multiplication by A^m, the upper-left entry is $ru + sme = 1$. If K is the lower-left entry, then

$$\begin{aligned} rK &= rm - rke + skemt \\ &= k(r(y - e) + smet) \\ &= kt(ru + sme) = kt. \end{aligned}$$

Thus

$$B^{ry} A^{-ke} B^{-t} A^{m} = \begin{pmatrix} 1 & se \\ kt/r & 1 + sket/r \end{pmatrix}.$$

Right multiplication by $B^{-re} = U(-se)$ proves (2.3). The matrices $L(kt/r)$ and A commute.

If m, y, t, e are nonzero, then $k \neq 0$, and all five exponents in $w = b^{ry} a^{-ke} b^{-t} a^{m} b^{-re}$ are nonzero. In $[w, a] = waw^{-1}a^{-1}$, the generator types alternate as $b, a, b, a, b, a, b, a, b, a, b, a$. Consequently the word is freely reduced and nonempty, while its matrix value is I. □

The next lemma is the good-class criterion [1, Definition 3.3 and Lemma 3.5], stated with the explicit relation length needed here. We include the proof, in particular its normalization when the denominator is not coprime to the numerator.

Lemma 2.2 (Good residue classes)**.** *Let s, w, m, y be nonzero integers, and put*

$$D = \gcd(w, sm) > 0, \qquad d = \gcd(w, s) > 0.$$

Suppose

$$yD \mid md, \qquad w \mid smy - D \text{ or } w \mid smy + D. \tag{2.4}$$

Every integer

$$r \equiv w \pmod{sm}, \qquad r \notin \{0, w, D, -D\},$$

admits a five-syllable lower-unipotent word for s/r, and therefore a nontrivial 12-syllable relation.

Proof. First consider the normalized situation

$$y \mid m, \qquad w \mid smy - 1, \qquad r = w + smt \notin \{0, w, 1, -1\}.$$

Set $u = (1 - smy)/w$, choose $k \in \mathbb{Z}$ with $m = ky$, and put $e = y - ut$. Then

$$ru + sme = (w + smt)u + sm(y - ut) = wu + smy = 1.$$

Here $t \neq 0$ because $r \neq w$. Also $e \neq 0$, since $e = 0$ would give $ru = 1$ and hence $r \in \{1, -1\}$. Apply Lemma 2.1. If instead $w \mid smy + 1$, replace y by $-y$. This changes neither divisibility by y nor its nonvanishing.

For the general case, put

$$c = D/d, \quad w' = w/D, \quad r' = r/D, \quad s' = s/d, \quad m' = md/D.$$

All these quantities are integers. Indeed, $D/d = \gcd(w/d, m)$ because $\gcd(w/d, s/d) = 1$, so $c \mid m$. Moreover, $D \mid r$ and

$$r' = w' + s'm't, \qquad y \mid m', \qquad w' \mid s'm'y - 1 \text{ or } w' \mid s'm'y + 1,$$

where $r = w + smt$ and $t \neq 0$. The exclusions on r give $r' \notin \{0, w', 1, -1\}$. The normalized argument produces a five-syllable word for s'/r', even when $s' = 1$. Finally,

$$\frac{s'}{r'} = c\frac{s}{r}, \qquad B_{s'/r'} = (B_{s/r})^c.$$

Multiply every b exponent by the nonzero integer c. The matrix identity, nonzero exponents, and alternating syllable pattern are preserved. □

3 The two remaining progressions

Proposition 3.1. *For every nonzero integer x, let $R = 1 + 1260x$ and $B = B_{24/R}$. Then*

$$B^{3Rx} A^{18+630x} B^{-3} A^{18x} B^{R(3+105x)} = L(18/R). \tag{3.1}$$

The commutator of the displayed formal word with a is a nontrivial 12-syllable relation.

Proof. Take

$$m = 18x, \quad y = 3x, \quad k = 6, \quad u = 1 + 36x, \quad t = 3, \quad e = -3 - 105x. \tag{3.2}$$

Then $m = ky$ and $e = y - ut$. A direct expansion gives

$$\begin{aligned} Ru + 24me &= (1 + 1260x)(1 + 36x) + 432x(-3 - 105x) \\ &= 1 + 1296x + 45360x^2 - 1296x - 45360x^2 = 1. \end{aligned}$$

Since x is a nonzero integer, $|R| \geq 1259$, and m, y, t are nonzero. The equality $e = 0$ would imply $35x = -1$, which is impossible in $\mathbb{Z}$. Thus Lemma 2.1 applies. Substituting (3.2) into (2.3) gives precisely (3.1), with $kt = 18$. □

Corollary 3.2. *If $r \equiv \pm 1 \pmod{1260}$ and $|r| > 1$, then $24/r$ has a five-syllable lower-unipotent word and a nontrivial 12-syllable relation. In particular, this holds for every integer whose residue modulo 40320 is in (1.3).*

Proof. Choose $\varepsilon \in \{1, -1\}$ such that $R = \varepsilon r$ is congruent to 1 modulo 1260. Write $R = 1 + 1260x$. Then $x \neq 0$, since $|r| > 1$. Apply Proposition 3.1. Because $B_{24/R} = B_{24/r}^{\varepsilon}$, multiply the three b exponents in (3.1) by ε. This does not change nonvanishing or syllable length. For the final assertion, $1260 \mid 40320$, and the four representatives in (1.3) reduce to $1, -1, 1, -1$ modulo 1260. □

For example, $R = 1261$ corresponds to $x = 1$, and

$$B_{24/1261}^{3783} A^{648} B_{24/1261}^{-3} A^{18} B_{24/1261}^{136188} = L(18/1261).$$

For $r = 6299$, choose $R = -6299$ and $x = -5$. In the original parameter $24/6299$, this becomes

$$B_{24/6299}^{-94485} A^{-3132} B_{24/6299}^{3} A^{-90} B_{24/6299}^{-3288078} = L(-18/6299).$$

The examples illustrate the sign convention; the proof of Proposition 3.1 applies to all nonzero integers x.

4 The finite certificate

Let $\mathcal{T}$ be the 48 quadruples (w, m, y, D) in Appendix A, and let $M = 40320$. Each row denotes both residue classes $r \equiv w$ and $r \equiv -w \pmod{24m}$. Define

$$\mathcal{C} = \bigcup_{(w,m,y,D)\in\mathcal{T}} \{a \in \{0, \dots, M-1\} : a \equiv \pm w \pmod{24m}\}. \tag{4.1}$$

Let

$$\begin{aligned} E = \{&7, 8, 9, 10, 11, 12, 13, 14, 17, 24, 31, 35, 37, 59, 61, 83, 85, \\ &107, 109, 613, 877, 1139, 2267\}. \end{aligned} \tag{4.2}$$

Proposition 4.1 (Finite covering certificate)**.** *The table $\mathcal{T}$ has the following properties.*

(i) *Every row satisfies $24m \mid M$, $D = \gcd(w, 24m)$, $yD \mid m \gcd(w, 24)$, and $w \mid 24my - D$ or $w \mid 24my + D$.*

(ii) *The complement of $\mathcal{C}$ is precisely $\{1261, 6299, 34021, 39059\}$.*

(iii) *The set of numbers greater than 6 occurring as a w or D in the table is exactly E.*

Proof. These are finite integer assertions about the complete table in Appendix A. For clarity, the entire covering check is the following enumeration: start with $\mathcal{C} = \varnothing$; for each row and each $0 \le a < M$, insert a if $24m$ divides $a - w$ or $a + w$; finally take the complement in $\{0, \dots, M-1\}$. The result is the four-element set in (ii). The rowwise divisibilities and the set in (iii) are checked by ordinary integer arithmetic. The accompanying file `verify_certificate.py` performs these exact checks from `data/compact_certificate.json`; `FiniteCover.lean` proves the same finite assertions using kernel-checked decision procedures. The tables and the stated tests fully specify the certificate, independently of how its rows were found. □

Lemma 4.2 (Exceptional words)**.** *For each $r \in E$, the exponent list in Appendix C defines an alternating word $w_r = b^{e_1} a^{e_2} \cdots b^{e_L}$ with nonzero exponents and $L \le 15$. Its value is*

$$W_r = \begin{pmatrix} \alpha & 0 \\ \beta & \alpha^{-1} \end{pmatrix}, \tag{4.3}$$

where the exact rational numbers α, β are given in Appendix D. Consequently $[w_r a w_r^{-1}, a]$ is a nontrivial relation with $4L + 4 \le 64$ syllables.

Proof. Multiply the matrices in (2.1) using each exponent list. For a direct scalar recurrence, if the accumulated matrix is $\left(\begin{smallmatrix} p & q \\ v & z \end{smallmatrix}\right)$, right multiplication by $B_{24/r}^n$ or A^n respectively gives

$$(p, q, v, z) \longmapsto (p, q + 24np/r, v, z + 24nv/r),$$

$$(p, q, v, z) \longmapsto (p + nq, q, v + nz, z).$$

Starting from $(1, 0, 0, 1)$ gives exactly the entries in Appendix D. This also specifies a short exact-rational verification of every row.

For any matrix of the form (4.3),

$$W_r A W_r^{-1} = L(\alpha^{-2}),$$

which commutes with A. The formal relation expands as

$$w_r a w_r^{-1} a \, w_r a^{-1} w_r^{-1} a^{-1}.$$

Each copy of w_r or w_r^{-1} begins and ends with a b syllable, and the connecting syllables are nonzero powers of a. There is no free cancellation, so its syllable length is exactly $4L + 4$. The table has $L \leq 15$. □

Proposition 4.3. *For every integer r with $|r| > 6$, there is a nontrivial relation for $A, B_{24/r}$ of at most 64 syllables.*

Proof. Since $B_{-q} = B_q^{-1}$, it suffices to handle $r > 6$. If $r \in E$, use Lemma 4.2. Otherwise, if its residue modulo M is one of (1.3), use Corollary 3.2. In the remaining case, Proposition 4.1(ii) supplies a row with $r \equiv \pm w \pmod{24m}$. Replace r by $-r$ if necessary to obtain the positive-sign class, apply Lemma 2.2, and then invert the b generator back. All excluded values $0, \pm w, \pm D$ with absolute value greater than 6 belong to $\pm E$ by Proposition 4.1(iii); our chosen denominator is therefore not excluded. The good-class and progression constructions have 12 syllables, and the exceptional words give at most 64. □

Proof of Corollary 1.2. Outside $\pm E$, the proof just given produces a five-syllable lower-unipotent word. For $r \in E$ with $r > 17$, every word listed in Appendix C has three or five syllables. These are precisely witnesses of the form (1.2). Negating the b exponents handles negative denominators. □

5 From relations to abstract nonfreeness

For completeness we give the group-theoretic bridge, including the Hopf property used in the formal development. This avoids identifying failure of freeness of the displayed generating pair with abstract nonfreeness without justification.

Lemma 5.1. *Every surjective endomorphism of a finitely generated free group is injective.*

Proof. First, a nonidentity element w in any free group is detected by a finite permutation representation. Choose a word spelling w, and let S be the finite set of its suffix products, including 1 and w. For each free generator g, left multiplication by g defines a partial bijection on S, on the domain where both the initial and final elements lie in S. Extend it to a permutation of S by matching the two equally sized complements. The inverse permutation agrees with multiplication by g^{-1} wherever both endpoints lie in S. By the universal property of the free group, these permutations

define a homomorphism to $\mathrm{Sym}(S)$. Reading the chosen word from right to left sends 1 successively through its suffix products and finally to $w \neq 1$. The homomorphism therefore detects w.

Now let F be finitely generated and let $\phi : F \to F$ be surjective. For any finite group K, the set $\mathrm{Hom}(F, K)$ is finite. Precomposition $f \mapsto f \circ \phi$ is injective, hence surjective, on this set. If a nonidentity w belonged to $\ker \phi$, choose a homomorphism $f : F \to K$ detecting it as in the first paragraph. Write $f = g \circ \phi$. Then $f(w) = g(1) = 1$, a contradiction. □

Lemma 5.2. *Let H be generated by two noncommuting elements A, B. If the evaluation homomorphism $F(a, b) \to H$ has nontrivial kernel, then H is not an abstract free group.*

Proof. Suppose that H were free. Its noncommutativity forces a free basis to contain at least two elements. Sending two distinct basis elements to a, b, and all other basis elements to 1, gives a surjection $H \to F(a, b)$. Composing this map with the surjective evaluation map $F(a, b) \to H$ gives a surjective endomorphism of $F(a, b)$. By Lemma 5.1 it is injective, forcing the evaluation map to be injective as well. This contradicts its nontrivial kernel. □

Proof of Theorem 1.1. The relation and its length bound follow from Proposition 4.3. For $q = 24/r \neq 0$,

$$AB_q = \begin{pmatrix} 1 & q \\ 1 & 1+q \end{pmatrix}, \qquad B_q A = \begin{pmatrix} 1+q & q \\ 1 & 1 \end{pmatrix},$$

so A and B_q do not commute. Apply Lemma 5.2. All matrices and identities already lie in $\mathrm{SL}_2(\mathbb{Q})$. The entrywise inclusion $\mathrm{SL}_2(\mathbb{Q}) \hookrightarrow \mathrm{SL}_2(\mathbb{C})$ is injective and identifies the generated subgroups, so the same abstract nonfreeness conclusion holds in $\mathrm{SL}_2(\mathbb{C})$. □

6 Formal statements and reproducibility

The accompanying project is pinned to Lean 4.24.0 and mathlib v4.24.0. The source file `Challenge.lean` specifies the displayed matrices and target propositions independently of the implementation; `Solution.lean` proves those propositions. The implementation uses the actual free group on a two-element type, with its standard nontriviality criterion. A syllable is a pair in $\mathrm{Bool} \times \mathbb{Z}$, with `false` denoting a and `true` denoting b. The reducedness predicate requires alternating generator types and nonzero exponents. Its connection to a nonidentity free-group element is proved by expansion into signed letters.

The central bounded-relation theorem has the mathematical content

$$\forall r \in \mathbb{Z}, \quad |r| > 6 \implies \exists \ell : \ell \text{ is nonempty and reduced}, \quad |\ell| \leq 64, \\ \mathrm{word}(\ell) \neq 1, \quad \mathrm{ev}_{24/r}(\mathrm{word}(\ell)) = I.$$

The separate nonfreeness theorem concludes $\neg\, \mathrm{IsFreeGroup}(G_{24/r})$. The rational-to-complex map is proved injective, and its image is proved to be exactly the subgroup generated by the complex versions of (1.1). Thus the complex conclusion is an explicit formal theorem.

Mathematical component	Lean source or declaration
Matrices and reduced words	`Core.lean`
General good-class criterion	`GoodRuleArithmetic.lean`, `GoodRule.lean`
Single-parameter identity	`SimpleDirect.lean`
Finite cover and exceptional words	`FiniteCover.lean`, `FiniteWords.lean`
Hopf property and nonfreeness	`FreeGroupHopf.lean`, `NonFree.lean`
Uniform syllable bound	`TwentyFour.twentyfour_boundedRelation`
Two-step conclusion for $\|r\| > 17$	`TwentyFour.twentyfour_twoStep`
Complex matrix-group conclusion	`TwentyFour.twentyfour_complex_not_free`

The single-parameter construction in Proposition 3.1 is proved in `SimpleDirect.lean`; the declaration `simple_direct_power_identity` states the five integer powers in (3.1) directly. Both the universal main proof and the bounded-relation proof use this construction.

In `TwoStep.lean`, the theorem `twentyfour_twoStep` separately proves Corollary 1.2. Its predicate requires a nonzero exponent list of length 1, 3, or 5, beginning with b, whose matrix has zero upper-right entry. This proof checks the triangular witness directly rather than inferring it from the 64-syllable relation bound.

The final source revision passed Lean verification against the independent specifications, an axiom-dependency audit, and independent nanoda replay. The replay includes nonfreeness over $\mathbb{Q}$ and $\mathbb{C}$, nontrivial relation existence, the 64-syllable bound, the two-step conclusion, and the direct parametric identity and certificates. The accompanying verification report records the exact target list and source and proof-export hashes.

The finite coverage proof checks 40 blocks of 1008 residues, which together exhaust $0, \ldots, 40319$. It uses Lean's ordinary `decide`, producing proof terms checked by the kernel. The project does not use `sorry`, user-added axioms, `native_decide`, or compiler-oracle axioms. The audited logical dependencies are restricted to Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. Independent nanoda replay checks exported declarations and their transitive dependencies under the same allowed-axiom list.

No acceptance by the full sandboxed Palomar/comparator workflow is claimed; compatibility and host limitations are documented in the repository. Version pins, commands, exact theorem lists, and raw checker logs are supplied there. The standalone Python verifier uses only integers and `fractions.Fraction`; it is a convenient second implementation of the finite arithmetic, rather than a trusted step in the Lean proof.

AI assistance

GPT-6-Astra assisted with research exploration, proof construction, exact computation, Lean formalization, and preparation of this manuscript.

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A Complete good-class table

Each row (w, m, y, D) represents the two classes $r \equiv \pm w \pmod{24m}$. The full arithmetic conditions are those in Proposition 4.1(i). The row numbers are for reference only; the machine-readable certificate contains the same ordered quadruples.

i	w	m	y	D	i	w	m	y	D
1	1	1	1	1	25	13	7	1	1
2	2	1	1	2	26	35	7	1	7
3	3	1	1	3	27	107	7	7	1
4	4	1	1	4	28	13	8	4	1
5	5	1	1	1	29	59	8	4	1
6	6	1	1	6	30	13	10	2	1
7	8	1	1	8	31	35	10	1	5
8	9	1	1	3	32	37	10	2	1
9	12	1	1	12	33	13	15	3	1
10	24	1	1	24	34	35	15	3	5
11	7	2	1	1	35	83	15	3	1
12	10	2	1	2	36	37	16	8	1
13	14	2	2	2	37	83	16	8	1
14	7	3	3	1	38	59	30	5	1
15	31	3	3	1	39	61	30	5	1
16	11	4	4	1	40	85	30	2	5
17	35	4	4	1	41	37	84	2	1
18	7	5	1	1	42	109	84	2	1
19	11	5	1	1	43	59	140	20	1
20	17	5	1	1	44	1139	140	20	1
21	11	6	1	1	45	83	280	28	1
22	13	6	1	1	46	2267	280	28	1
23	17	6	2	1	47	613	560	40	1
24	11	7	7	1	48	877	560	40	1

B Exact finite verification

Given the table above, the finite part can be checked as follows.

1. For each row, compute the two gcd values and check the gcd and divisibility conditions of Proposition 4.1(i).
2. Evaluate the predicate in (4.1) for every integer a from 0 to 40319. Its false values must be exactly $1261, 6299, 34021, 39059$.
3. Collect the values w, D exceeding 6, remove repetitions, and compare with (4.2).
4. For every word in Appendix C, apply the rational recurrences in Lemma 4.2 and compare the resulting matrix with Appendix D.

The supplied verifier also checks the full commutator matrices, reduced syllable patterns, and the bound 64. It regenerates all three tables from the checked data, so the exponent lists and matrix entries need not be copied manually.

C Complete exceptional exponent lists

For each listed denominator r, put $B = B_{24/r}$. The list $(e_1, \dots, e_L)$ denotes

$$W_r = B^{e_1} A^{e_2} B^{e_3} \cdots B^{e_L}.$$

All exponents are nonzero, and every list has odd length. The relation used uniformly is $[W_r A W_r^{-1}, A] = I$, with $4L + 4$ syllables. For $r > 17$ all these lists have length at most five, as required for Corollary 1.2.

r	$(e_1, \dots, e_L)$	L
7	$(1, -1, 1, -1, 1, -1, 1, -5, -1, 1, -1, 21, -1, 1, -1)$	15
8	$(1, -1, 1, -1, 1)$	5
9	$(1, -1, 1, -1, 6)$	5
10	$(1, -1, 1, -1, -1, 5, -1)$	7
11	$(1, -1, 1, -3, 22)$	5
12	$(1, -1, 1)$	3
13	$(1, -1, 1, 3, -26)$	5
14	$(1, -1, 1, 1, 1, 2, 4, 3, -7)$	9
17	$(15, -1, 1, -3, 3, -19, 4, 8, -119)$	9
24	$(2, -1, 2)$	3
31	$(1, -1, -4, -3, -93)$	5
35	$(2, -1, 5, 4, -7)$	5
37	$(3, -1, 3, 9, -222)$	5
59	$(2, -1, -11, 9, 177)$	5
61	$(4, -1, 7, -89, 61)$	5
83	$(2, -2, 13, -15, 249)$	5
85	$(2, -2, 15, 8, -17)$	5
107	$(4, -1, -39, 49, 107)$	5
109	$(1, -5, 10, -50, 109)$	5
613	$(25, -1, -1179, 166, 1226)$	5
877	$(2, -18, -135, 475, 877)$	5
1139	$(8, -6, 701, -617, 1139)$	5
2267	$(1, -94, -206, -1039, -2267)$	5

D Exact matrices for the exceptional words

The values of the words in Appendix C are $W_r = \left(\begin{smallmatrix} \alpha & 0 \\ \beta & \alpha^{-1} \end{smallmatrix}\right)$. The last column is the length of the displayed commutator relation; no optimality of these lengths is asserted.

r	α	β	$4L+4$
7	$\frac{1}{2401}$	$-\frac{410055}{343}$	64
8	1	1	24
9	$\frac{1}{9}$	$\frac{2}{3}$	24
10	$\frac{1}{25}$	$-\frac{57}{5}$	32
11	$\frac{1}{121}$	$\frac{28}{11}$	24
12	-1	-1	16
13	$\frac{1}{169}$	$-\frac{46}{13}$	24
14	$\frac{1}{2401}$	$-\frac{70501}{343}$	40
17	$\frac{1}{83521}$	$-\frac{2444271}{4913}$	40
24	-1	-1	16
31	$\frac{1}{961}$	$-\frac{412}{31}$	24
35	$\frac{1}{49}$	$-\frac{75}{7}$	24
37	$\frac{1}{1369}$	$-\frac{352}{37}$	24
59	$\frac{1}{3481}$	$\frac{2848}{59}$	24
61	$\frac{1}{3721}$	$\frac{9462}{61}$	24
83	$\frac{1}{6889}$	$\frac{7949}{83}$	24
85	$\frac{1}{289}$	$-\frac{1050}{17}$	24
107	$\frac{1}{11449}$	$\frac{51000}{107}$	24
109	$\frac{1}{11881}$	$\frac{54005}{109}$	24
613	$\frac{1}{375769}$	$\frac{4798281}{613}$	24
877	$\frac{1}{769129}$	$\frac{28102789}{877}$	24
1139	$\frac{1}{1297321}$	$\frac{61572851}{1139}$	24
2267	$\frac{1}{5139289}$	$-\frac{485429215}{2267}$	24

References

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